

Task 3: Develop a C program using fork() that creates a parent and child process. Display the Process ID (PID), Parent Process ID (PPID), and process states at different stages of execution.

Flow Chart



```
GNU nano 7.2 practical3_fork.c *
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <stdlib.h>

int main()
{
    pid_t pid;

    printf("Before fork()\n");
    printf("Parent PID: %d\n", getpid());

    pid = fork();

    if (pid < 0)
    {
        perror("fork failed");
        return 1;
    }

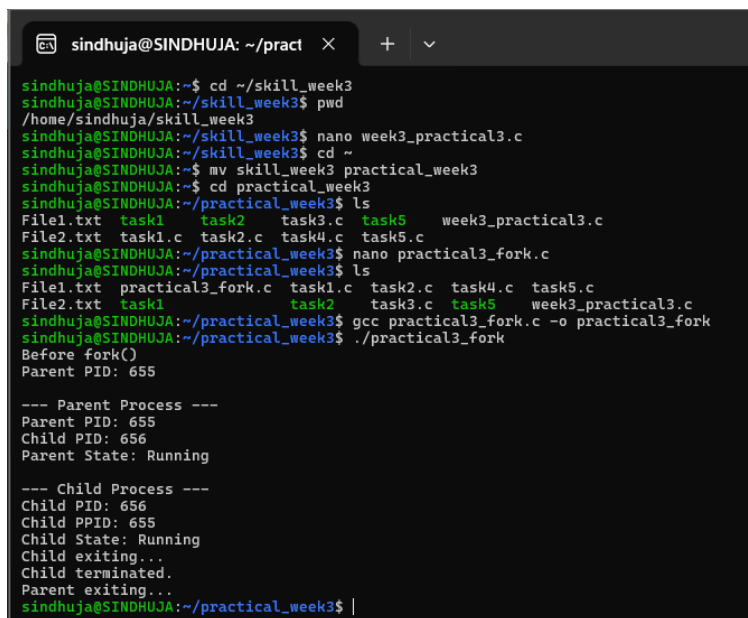
    if (pid == 0)
    {
        printf("\n--- Child Process ---\n");
        printf("Child PID: %d\n", getpid());
        printf("Child PPID: %d\n", getppid());
        printf("Child State: Running\n");

        sleep(5);

        printf("Child exiting...\n");
        exit(0);
    }
    else
    {
        printf("\n--- Parent Process ---\n");
        printf("Parent PID: %d\n", getpid());
        printf("Child PID: %d\n", pid);
        printf("Parent State: Running\n");

        wait(NULL);

        printf("Child terminated.\n");
        printf("Parent exiting...\n");
    }
}
```



```
sindhuja@SINDHUJA: ~/pract
sindhuja@SINDHUJA:~$ cd ~/skill_week3
sindhuja@SINDHUJA:~/skill_week3$ pwd
/home/sindhuja/skill_week3
sindhuja@SINDHUJA:~/skill_week3$ nano week3_practical3.c
sindhuja@SINDHUJA:~/skill_week3$ cd ~
sindhuja@SINDHUJA:~$ mv skill_week3 practical_week3
sindhuja@SINDHUJA:~$ cd practical_week3
sindhuja@SINDHUJA:~/practical_week3$ ls
File1.txt task1 task2 task3.c task5 week3_practical3.c
File2.txt task1.c task2.c task4.c task5.c
sindhuja@SINDHUJA:~/practical_week3$ nano practical3_fork.c
sindhuja@SINDHUJA:~/practical_week3$ ls
File1.txt practical3_fork.c task1.c task2.c task4.c task5.c
File2.txt task1 task2 task3.c task5 week3_practical3.c
sindhuja@SINDHUJA:~/practical_week3$ gcc practical3_fork.c -o practical3_fork
sindhuja@SINDHUJA:~/practical_week3$ ./practical3_fork
Before fork()
Parent PID: 655

--- Parent Process ---
Parent PID: 655
Child PID: 656
Parent State: Running

--- Child Process ---
Child PID: 656
Child PPID: 655
Child State: Running
Child exiting...
Child terminated.
Parent exiting...
sindhuja@SINDHUJA:~/practical_week3$ |
```

```

sindhuja@SINDHUJA:~/practical_week3$ ./practical3_fork
Before fork()
Parent PID: 745

--- Parent Process ---
Parent PID: 745
Child PID: 746
Parent State: Running

--- Child Process ---
Child PID: 746
Child PPID: 745
Child State: Running
Child exiting...
Child terminated.
Parent exiting...

```

```

sindhuja@SINDHUJA:~/practical_week3$ Before fork()
Parent PID: 753

--- Parent Process ---
Parent PID: 753
Child PID: 754
Parent State: Running

--- Child Process ---
Child PID: 754
Child PPID: 753
Child State: Running
Child exiting...
Child terminated.
Parent exiting...

[1]+  Done                  ./practical3_fork
sindhuja@SINDHUJA:~/practical_week3$ ps
  PID TTY          TIME CMD
 297 pts/0    00:00:00 bash
 757 pts/0    00:00:00 ps
sindhuja@SINDHUJA:~/practical_week3$ ps -o pid,ppid,state,cmd
  PID  PPID S  CMD
 297    296 S  -bash
 758    297 R  ps -o pid,ppid,state,cmd
sindhuja@SINDHUJA:~/practical_week3$ cat /proc/2501/status
cat: /proc/2501/status: No such file or directory

```

```

sindhuja@SINDHUJA:~/practical_week3$ Before fork()
Parent PID: 753

--- Parent Process ---
Parent PID: 753
Child PID: 754
Parent State: Running

--- Child Process ---
Child PID: 754
Child PPID: 753
Child State: Running
Child exiting...
Child terminated.
Parent exiting...

[1]+  Done                  ./practical3_fork
sindhuja@SINDHUJA:~/practical_week3$ ps
  PID TTY          TIME CMD
 297 pts/0    00:00:00 bash
 757 pts/0    00:00:00 ps
sindhuja@SINDHUJA:~/practical_week3$ ps -o pid,ppid,state,cmd
  PID  PPID S  CMD
 297    296 S  -bash
 758    297 R  ps -o pid,ppid,state,cmd
sindhuja@SINDHUJA:~/practical_week3$ cat /proc/2501/status
cat: /proc/2501/status: No such file or directory

```



sindhuja@SINDHUJA: ~/pract



```
top - 13:33:03 up 41 min, 1 user, load average: 0.00, 0.02, 0.01
Tasks: 22 total, 1 running, 21 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7782.6 total, 7000.9 free, 513.3 used, 420.1 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used. 7269.3 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	21524	13056	9732	S	0.0	0.2	0:00.56	systemd
2	root	20	0	3180	2212	2072	S	0.0	0.0	0:00.00	init-systemd(Ub
6	root	20	0	3196	2124	2008	S	0.0	0.0	0:00.00	init
52	root	19	-1	50456	16388	15228	S	0.0	0.2	0:00.24	systemd-journal
97	root	20	0	25080	6516	5156	S	0.0	0.1	0:00.60	systemd-udev
166	systemd+	20	0	21460	13188	10932	S	0.0	0.2	0:00.12	systemd-resolve
167	systemd+	20	0	91028	7956	6988	S	0.0	0.1	0:00.18	systemd-timesyn
176	root	20	0	4244	2684	2432	S	0.0	0.0	0:00.00	cron
177	message+	20	0	9632	5412	4716	S	0.0	0.1	0:00.10	dbus-daemon
184	root	20	0	17972	8888	7848	S	0.0	0.1	0:00.08	systemd-logind
187	root	20	0	1755848	11392	8980	S	0.0	0.1	0:00.24	wsl-pro-service
195	root	20	0	3124	1972	1836	S	0.0	0.0	0:00.00	agetty
196	syslog	20	0	222516	6020	4616	S	0.0	0.1	0:00.10	rsyslogd
213	root	20	0	107020	23132	13672	S	0.0	0.3	0:00.10	unattended-upgr
295	root	20	0	3184	1116	980	S	0.0	0.0	0:00.00	SessionLeader
296	root	20	0	3200	1252	1108	S	0.0	0.0	0:00.02	Relay(297)
297	sindhuja	20	0	6592	5924	3668	S	0.0	0.1	0:00.04	bash
298	root	20	0	6700	4660	3876	S	0.0	0.1	0:00.00	login
341	sindhuja	20	0	20108	11448	9516	S	0.0	0.1	0:00.07	systemd
342	sindhuja	20	0	21156	3576	1852	S	0.0	0.0	0:00.00	(sd-pam)
369	sindhuja	20	0	6080	5324	3660	S	0.0	0.1	0:00.01	bash
762	sindhuja	20	0	9344	5740	3512	R	0.0	0.1	0:00.00	top

```
sindhuja@SINDHUJA:~/practical_week3$ ./practical3_fork
```

```
Before fork()
```

```
Parent PID: 770
```

```
--- Parent Process ---
```

```
Parent PID: 770
```

```
Child PID: 771
```

```
Parent State: Running
```

```
--- Child Process ---
```

```
Child PID: 771
```

```
Child PPID: 770
```

```
Child State: Running
```

```
Child exiting...
```

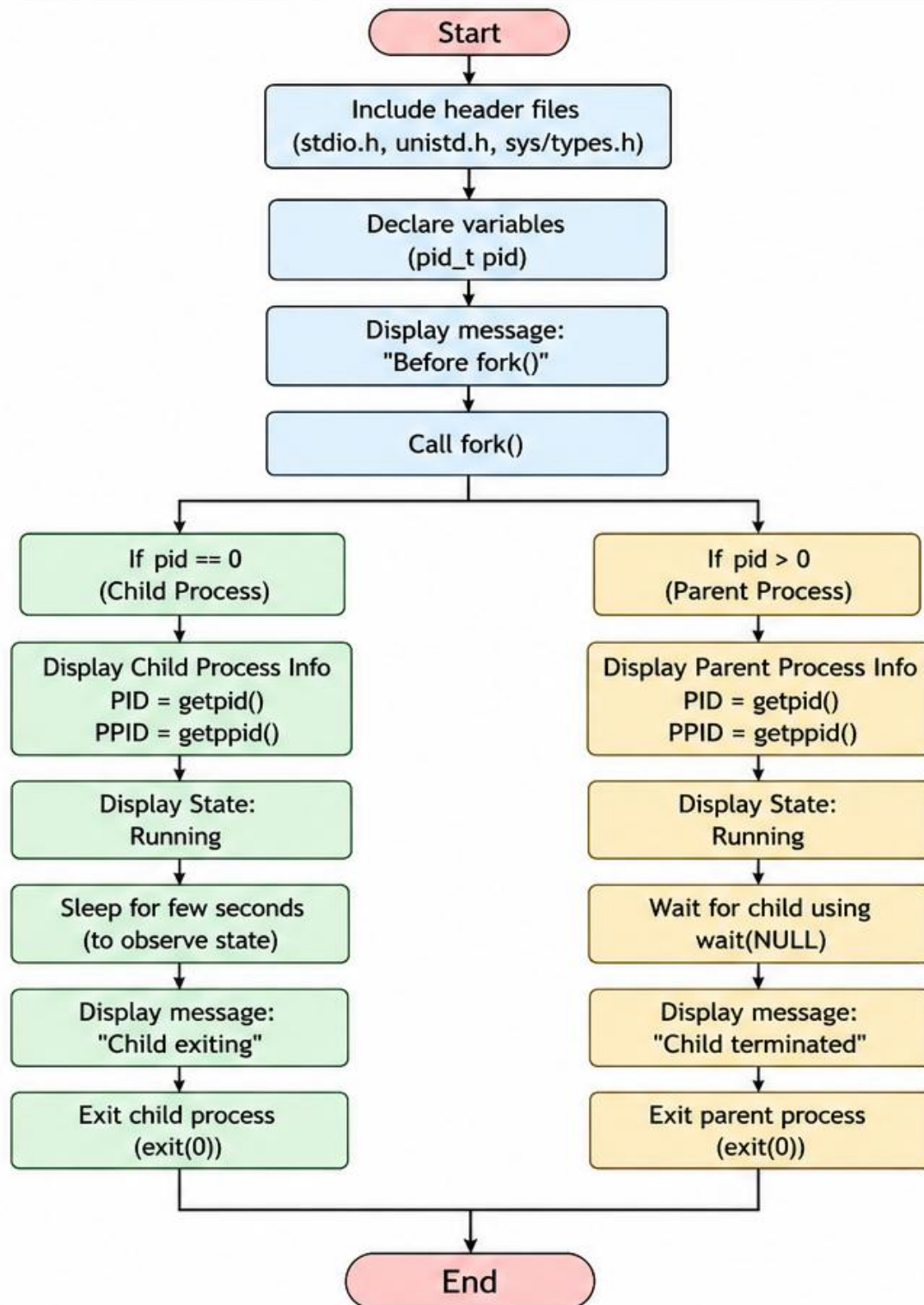
```
Child terminated.
```

```
Parent exiting...
```

```
sindhuja@SINDHUJA:~/practical_week3$
```

```
sindhuja@SINDHUJA:~/practical_week3$
```

Flowchart 1: C Program Using fork()



Task 4: Design an experiment to observe process state transitions (Ready, Running, Waiting, Terminated) using Linux monitoring tools such as ps, top, and /proc. Document the observations.

Flow Chart:

```
sindhuja@SINDHUJA:~/practical_week3$ kill 12345
-bash: kill: (12345) - No such process
[1]+  Done                  sleep 60
sindhuja@SINDHUJA:~/practical_week3$ ps -p 12345
PID TTY          TIME CMD
sindhuja@SINDHUJA:~/practical_week3$ cd ~/practical_week3
sindhuja@SINDHUJA:~/practical_week3$ sleep 60 &
[1] 790
sindhuja@SINDHUJA:~/practical_week3$ ps -p 4821 -o pid,ppid,state,cmd
PID PPID S  CMD
sindhuja@SINDHUJA:~/practical_week3$ cat /proc/4821/status
cat: /proc/4821/status: No such file or directory
sindhuja@SINDHUJA:~/practical_week3$ top
top - 13:42:51 up 51 min, 1 user, load average: 0.07, 0.02, 0.00
Tasks: 22 total, 1 running, 21 sleeping, 0 stopped, 0 zombie
%Cpu(s):  0.8 us,  0.0 sy,  0.0 ni, 99.2 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem : 7782.6 total, 6991.8 free, 522.2 used, 420.5 buff/cache
MiB Swap: 2048.0 total, 2048.0 free,  0.0 used. 7260.4 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	21524	13056	9732	S	0.0	0.2	0:00.62	systemd
2	root	20	0	3180	2212	2072	S	0.0	0.0	0:00.00	init-systemd(Ub
6	root	20	0	3196	2124	2008	S	0.0	0.0	0:00.00	init
52	root	19	-1	50456	16404	15244	S	0.0	0.2	0:00.30	systemd-journal
97	root	20	0	25080	6516	5156	S	0.0	0.1	0:00.66	systemd-udev
166	systemd+	20	0	21460	13188	10932	S	0.0	0.2	0:00.13	systemd-resolve
167	systemd+	20	0	91028	7956	6988	S	0.0	0.1	0:00.24	systemd-timesyn
176	root	20	0	4244	2684	2432	S	0.0	0.0	0:00.00	cron
177	message+	20	0	9632	5412	4716	S	0.0	0.1	0:00.12	dbus-daemon
184	root	20	0	17972	8888	7848	S	0.0	0.1	0:00.08	systemd-logind
187	root	20	0	1755848	11480	8980	S	0.0	0.1	0:00.26	wsl-pro-service
195	root	20	0	3124	1972	1836	S	0.0	0.0	0:00.00	agetty
196	syslog	20	0	222516	6020	4616	S	0.0	0.1	0:00.11	rsyslogd
213	root	20	0	107020	23132	13672	S	0.0	0.3	0:00.10	unattended-upgr
295	root	20	0	3184	1116	980	S	0.0	0.0	0:00.00	SessionLeader
296	root	20	0	3200	1252	1108	S	0.0	0.0	0:00.04	Relay(297)
297	sindhuja	20	0	6592	5924	3668	S	0.0	0.1	0:00.07	bash
298	root	20	0	6700	4660	3876	S	0.0	0.1	0:00.00	login
341	sindhuja	20	0	20108	11448	9516	S	0.0	0.1	0:00.08	systemd
342	sindhuja	20	0	21156	3576	1852	S	0.0	0.0	0:00.00	(sd-pam)
369	sindhuja	20	0	6080	5324	3660	S	0.0	0.1	0:00.01	bash
799	sindhuja	20	0	9344	5480	3256	R	0.0	0.1	0:00.00	top

```
sindhuja@SINDHUJA:~/practical_week3$ sleep 60 &
[1] 779
sindhuja@SINDHUJA:~/practical_week3$ ps -o pid,ppid,state,cmd
PID PPID S  CMD
297 296 S  -bash
779 297 S  sleep 60
782 297 R  ps -o pid,ppid,state,cmd
sindhuja@SINDHUJA:~/practical_week3$ cat /proc/12345/status
cat: /proc/12345/status: No such file or directory
sindhuja@SINDHUJA:~/practical_week3$ top
top - 13:39:38 up 48 min, 1 user, load average: 0.02, 0.03, 0.00
Tasks: 23 total, 1 running, 22 sleeping, 0 stopped, 0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem : 7782.6 total, 6991.5 free, 522.5 used, 420.4 buff/cache
MiB Swap: 2048.0 total, 2048.0 free,  0.0 used. 7260.1 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	21524	13056	9732	S	0.0	0.2	0:00.59	systemd
2	root	20	0	3180	2212	2072	S	0.0	0.0	0:00.00	init-systemd(Ub
6	root	20	0	3196	2124	2008	S	0.0	0.0	0:00.00	init
52	root	19	-1	50456	16400	15240	S	0.0	0.2	0:00.26	systemd-journal
97	root	20	0	25080	6516	5156	S	0.0	0.1	0:00.63	systemd-udev
166	systemd+	20	0	21460	13188	10932	S	0.0	0.2	0:00.12	systemd-resolve
167	systemd+	20	0	91028	7956	6988	S	0.0	0.1	0:00.19	systemd-timesyn
176	root	20	0	4244	2684	2432	S	0.0	0.0	0:00.00	cron
177	message+	20	0	9632	5412	4716	S	0.0	0.1	0:00.11	dbus-daemon
184	root	20	0	17972	8888	7848	S	0.0	0.1	0:00.08	systemd-logind
187	root	20	0	1755848	11444	8980	S	0.0	0.1	0:00.26	wsl-pro-service
195	root	20	0	3124	1972	1836	S	0.0	0.0	0:00.00	agetty
196	syslog	20	0	222516	6020	4616	S	0.0	0.1	0:00.11	rsyslogd
213	root	20	0	107020	23132	13672	S	0.0	0.3	0:00.10	unattended-upgr
295	root	20	0	3184	1116	980	S	0.0	0.0	0:00.00	SessionLeader
296	root	20	0	3200	1252	1108	S	0.0	0.0	0:00.03	Relay(297)
297	sindhuja	20	0	6592	5924	3668	S	0.0	0.1	0:00.05	bash
298	root	20	0	6700	4660	3876	S	0.0	0.1	0:00.00	login
341	sindhuja	20	0	20108	11448	9516	S	0.0	0.1	0:00.07	systemd
342	sindhuja	20	0	21156	3576	1852	S	0.0	0.0	0:00.00	(sd-pam)
369	sindhuja	20	0	6080	5324	3660	S	0.0	0.1	0:00.01	bash
779	sindhuja	20	0	3132	1904	1792	S	0.0	0.0	0:00.00	sleep
784	sindhuja	20	0	9344	5736	3508	R	0.0	0.1	0:00.00	top

```

sindhuj@SINDHUJA:~/pract × + v

[1]+  Done                  sleep 60
sindhuj@SINDHUJA:~/practical_week3$ kill 4821
-bash: kill: (4821) - No such process
sindhuj@SINDHUJA:~/practical_week3$ ps -p 4821
  PID TTY          TIME CMD
sindhuj@SINDHUJA:~/practical_week3$ sleep 60 &
[1] 881
sindhuj@SINDHUJA:~/practical_week3$ ps -p 881 -o pid,ppid,state,cmd
  PID    PPID  STATE CMD
   881     0  S    sleep 60
sindhuj@SINDHUJA:~/practical_week3$ cat /proc/881/status
Name: sleep
Umask: 0022
State: S (sleeping)
Tgid: 881
Ngid: 0
Pid: 881
PPid: 297
TracerPid: 0
Uid: 1000    1000    1000    1000
Gid: 1000    1000    1000    1000
FSDSize: 256
Groups: 27 100 1000
NSTgid: 881
NSpid: 881
NSgid: 881
NSSid: 297
Kthreadd: 0
VmPeak: 3132 kB
VmSize: 3132 kB
VmLck: 0 kB
VmPin: 0 kB
VmHWM: 1908 kB
VmRSS: 1908 kB
RssAnon:          116 kB
RssFile:         1792 kB
RssShmem:          0 kB
VmData: 224 kB
VmStk: 132 kB
VmExe: 16 kB
VmLib: 1748 kB
VmPTE: 48 kB
VmSwap: 0 kB
HugetlbPages: 0 kB
CoreDumping: 0
THP_enabled: 1
untag_mask: 0xffffffffffffffff

```

```
cat /proc/self/status | grep -E "Name|State|Pid|PPid" | sed -e 's/^//; s/:$//'
```

```
Threads:      1  
SigQ:         1/31087  
SigPnd:       0000000000000000  
ShdPnd:       0000000000000000  
SigBlk:       0000000000000000  
SigIgn:       0000000000000000  
SigCgt:       0000000000000000  
CapInh:       0000000000000000  
CapPrm:       0000000000000000  
CapEff:       0000000000000000  
CapBnd:       000001fffffffffffff  
CapAmb:       0000000000000000  
NoNewPrivs:   0  
Seccomp:      0  
Seccomp_filters: 0  
Speculation_Store_Bypass: thread vulnerable  
SpeculationIndirectBranch: conditional enabled  
Cpus_allowed:    fff  
Cpus_allowed_list: 0-11  
Mems_allowed:     00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,  
00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,  
000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,00000000,  
Mems_allowed_list: 0  
voluntary_ctxt_switches: 1  
nonvoluntary_ctxt_switches: 0  
sindhuja@SINDHUJA:~/practical_week3$ grep -E "Name|State|Pid|PPid" /proc/801/status  
Name:        sleep  
State:       S (sleeping)  
Pid:         801  
PPid:        297  
TracerPid:   0
```

```

TracerPid: 0
sindhuja@SINDHUJA:~/practical_week3$ top
top - 13:44:16 up 53 min, 1 user, load average: 0.01, 0.01, 0.00
Tasks: 23 total, 1 running, 22 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.8 sy, 0.0 ni, 99.2 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7782.6 total, 6989.7 free, 524.3 used, 420.5 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7258.3 avail Mem

```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	21524	13056	9732	S	0.0	0.2	0:00.62	systemd
2	root	20	0	3180	2212	2072	S	0.0	0.0	0:00.00	init-systemd(Ub
6	root	20	0	3196	2124	2008	S	0.0	0.0	0:00.00	init
52	root	19	-1	50456	16408	15248	S	0.0	0.2	0:00.30	systemd-journal
97	root	20	0	25080	6516	5156	S	0.0	0.1	0:00.66	systemd-udev
166	systemd+	20	0	21460	13188	10932	S	0.0	0.2	0:00.13	systemd-resolve
167	systemd+	20	0	91028	7956	6988	S	0.0	0.1	0:00.24	systemd-timesyn
176	root	20	0	4244	2684	2432	S	0.0	0.0	0:00.00	cron
177	message+	20	0	9632	5412	4716	S	0.0	0.1	0:00.12	dbus-daemon
184	root	20	0	17972	8888	7848	S	0.0	0.1	0:00.08	systemd-logind
187	root	20	0	1755848	11496	8980	S	0.0	0.1	0:00.28	wsl-pro-service
195	root	20	0	3124	1972	1836	S	0.0	0.0	0:00.00	agetty
196	syslog	20	0	222516	6020	4616	S	0.0	0.1	0:00.11	rsyslogd
213	root	20	0	107020	23132	13672	S	0.0	0.3	0:00.10	unattended-upgr
295	root	20	0	3184	1116	980	S	0.0	0.0	0:00.00	SessionLeader
296	root	20	0	3200	1252	1108	S	0.0	0.0	0:00.04	Relay(297)
297	sindhuja	20	0	6592	5924	3668	S	0.0	0.1	0:00.08	bash
298	root	20	0	6700	4660	3876	S	0.0	0.1	0:00.00	login
341	sindhuja	20	0	20108	11448	9516	S	0.0	0.1	0:00.08	systemd
342	sindhuja	20	0	21156	3576	1852	S	0.0	0.0	0:00.00	(sd-pam)
369	sindhuja	20	0	6080	5324	3660	S	0.0	0.1	0:00.01	bash
801	sindhuja	20	0	3132	1908	1792	S	0.0	0.0	0:00.00	sleep
806	sindhuja	20	0	9344	5480	3256	R	0.0	0.1	0:00.00	top

```

[1]+  Done                  sleep 60
sindhuja@SINDHUJA:~/practical_week3$ kill 801
-bash: kill: (801) - No such process
sindhuja@SINDHUJA:~/practical_week3$ ps -p 801
  PID TTY          TIME CMD
sindhuja@SINDHUJA:~/practical_week3$ grep -E "Name|State|Pid|PPid" /proc/801/status
grep: /proc/801/status: No such file or directory
sindhuja@SINDHUJA:~/practical_week3$ sleep 300 &
[1] 835
sindhuja@SINDHUJA:~/practical_week3$ ps -p 835 -o pid,ppid,state,cmd
  PID  PPID S  CMD
  835   297 S  sleep 300
sindhuja@SINDHUJA:~/practical_week3$ grep -E "Name|State|Pid|PPid" /proc/835/status
Name: sleep
State: S (sleeping)
Pid: 835
PPid: 297
TracerPid: 0
sindhuja@SINDHUJA:~/practical_week3$ top
top - 14:16:37 up 59 min, 1 user, load average: 0.00, 0.00, 0.00
Tasks: 23 total, 1 running, 22 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7782.6 total, 6990.0 free, 521.1 used, 423.4 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7261.6 avail Mem

```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	21524	13056	9732	S	0.0	0.2	0:00.65	systemd
2	root	20	0	3180	2212	2072	S	0.0	0.0	0:00.00	init-systemd(Ub
6	root	20	0	3196	2124	2008	S	0.0	0.0	0:00.00	init
52	root	19	-1	50456	16420	15260	S	0.0	0.2	0:00.37	systemd-journal
97	root	20	0	25080	6516	5156	S	0.0	0.1	0:00.80	systemd-udev
166	systemd+	20	0	21460	13188	10932	S	0.0	0.2	0:00.13	systemd-resolve
167	systemd+	20	0	91028	7956	6988	S	0.0	0.1	0:00.26	systemd-timesyn
176	root	20	0	4244	2684	2432	S	0.0	0.0	0:00.00	cron
177	message+	20	0	9632	5412	4716	S	0.0	0.1	0:00.14	dbus-daemon
184	root	20	0	17972	8888	7848	S	0.0	0.1	0:00.09	systemd-logind
187	root	20	0	1755848	11580	9044	S	0.0	0.1	0:00.31	wsl-pro-service
195	root	20	0	3124	1972	1836	S	0.0	0.0	0:00.00	agetty
196	syslog	20	0	222516	6024	4616	S	0.0	0.1	0:00.11	rsyslogd
213	root	20	0	107020	23132	13672	S	0.0	0.3	0:00.10	unattended-upgr
295	root	20	0	3184	1116	980	S	0.0	0.0	0:00.00	SessionLeader
296	root	20	0	3200	1252	1108	S	0.0	0.0	0:00.07	Relay(297)
297	sindhuja	20	0	6592	5932	3668	S	0.0	0.1	0:00.10	bash
298	root	20	0	6700	4660	3876	S	0.0	0.1	0:00.00	login
341	sindhuja	20	0	20108	11448	9516	S	0.0	0.1	0:00.08	systemd
342	sindhuja	20	0	21156	3576	1852	S	0.0	0.0	0:00.00	(sd-pam)
369	sindhuja	20	0	6080	5324	3660	S	0.0	0.1	0:00.01	bash
835	sindhuja	20	0	3132	1908	1792	S	0.0	0.0	0:00.00	sleep
844	sindhuja	20	0	9344	5744	3512	R	0.0	0.1	0:00.00	top

```

sindhuja@SINDHUJA:~/practical_week3$ kill 835
sindhuja@SINDHUJA:~/practical_week3$ ps -p 835
  PID TTY          TIME CMD
[1]+  Terminated                  sleep 300
sindhuja@SINDHUJA:~/practical_week3$ ps -p 835 -o pid,ppid,state,cmd
  PID  PPID  S  CMD
sindhuja@SINDHUJA:~/practical_week3$

```

Flowchart 2: Experiment to Observe Process State Transitions

